

Biological and Psychological Effects of Physique Contest Preparation

BIOLOGICAL AND PSYCHOLOGICAL EFFECTS OF PHYSIQUE CONTEST PREPARATION

Reference	Participants	Biological Effects	Psychological Effects
Rossow et al. (2013)	N = 1 male, natural bodybuilder	- Blood pressure, testosterone, T3, and T4 decreased but recovered quickly - Ghrelin and cortisol levels increased	Mood disturbance
Kleiner et al. (1990)	N = 27 (70% male)	- Plasma glucose and magnesium decreased - Hemoglobin levels increased - Menstrual irregularities (25% of women)	
Kistler et al. (2014)	N = 1 male, natural bodybuilder	- Bone mineral content and density increased - Blood pressure and arterial stiffness decreased	
Trexler et al. (2017)	N = 15 (47% male)	- Cortisol increased but recovered - Testosterone recovered later - Ghrelin increased - Leptin increased - Insulin unchanged	
Hulmi et al. (2016)	N = 50 (all female)	- Leptin, T3, testosterone, estrogen, bone mass decreased but mostly recovered - Heart rate decreased - Menstrual irregularities	Mood changes (decreased vigor) but recovered
Halliday et al. (2016)	N = 1 female, natural bodybuilder	- Energy availability decreased - Loss of menses (recovered 71 weeks post-competition)	
Anderson et al. (1994)	N = 49 (all males)		- Binge eating frequency - Food preoccupation - Depression, anxiety, anger
Walberg-Rankin et al. (1993)	N = 6 (all female)	- Vitamin deficiency - Menstrual irregularities - Estradiol and progesterone decreased but recovered	
Robinson et al. (2015)	N = 1 male	Resting metabolic rate decreased	- No mood changes - No binge eating
Rohrig et al. (2017)	N = 1 female	- Leptin and thyroid stimulating hormone decreased - Estradiol and cortisol unchanged	- Energy decreased - Tiredness increased

Adapted from Helms et al. (2019)